Toward a four-dimensional Yang–Mills construction

Notebook 12: fine regularity on the flat current domain

Research working document

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Abstract

We strengthen the flat fine inverse from physical-cell Hilbert bounds to pointwise regularity for arbitrary bounded fine forcing. A discrete heat-kernel argument and the canceled finite-rank formula give uniform $C^{1,\alpha}$ estimates for every $\alpha < 1$, exponential cell locality, and a componentwise Laplacian bound at zero current. An explicit counterexample rules out the endpoint and individual second-difference bounds. Scalar gauge identities then place the full current perturbation in the actual homogeneous source norms N and N_β, without differentiating the current. A smaller uniform complex domain supports the full flat inverse and extension on bounded forcing. A massive heat-kernel subtraction also proves right-divergence estimates on Hölder data. Splitting the time integral at the output separation removes the exponent loss for each fixed exponent strictly between zero and one. A separate scalar proof applies under an explicit positive-penalty hypothesis. The distinct constrained-current inverse has the same bounded-forcing regularity on a further common ball. Nonflat backgrounds and retained histories remain separate.

Status. Research proofs checked by separate internal derivations and audits. No external certification or novelty claim is made. The baseline is valid for every odd M. The nonlinear constraint-current term uses $M = L^j$ with fixed odd $L \geq 3$, as proved in Notebook 11. All norms use physical fine derivatives and volume, at final block scale one.

1 Previously established algebraic inputs

We use the exact homogeneous inverse and scalar source definitions of Notebook 09, and the corrected quadratic-composition bound of Notebook 11. Let D be the fine scalar gradient, $\mathcal{L} = D^*D$, Q_0 the normalized scalar block average, and $Q = Q_M^{\mathrm{st}}$ the normalized straight one-form average. The scalar gauge is

$$G' = (\mathcal{L} + Q_0^* a' Q_0)^{-1}, \qquad R = I - G'Q_0^*(Q_0 G'^2 Q_0^*)^{-1} Q_0 G', \qquad Z = G'RG'.$$

The distinct positive penalty a enters the fine one-form inverse. The scalar penalty a' does not replace it. The identities

$$Q_0 Z = 0, \quad \mathcal{L}G'R = R, \quad Z\mathcal{L} = G'R, \quad \mathcal{L}Z\mathcal{L} = R$$

were proved from these definitions. They will also justify the fixed-refinement operator products on bounded forcing.

The canceled Bloch formulas $G_0 = F + \beta E_a^{-1}\beta^-$, including every central entry, are those proved in Notebook 09. We reuse their positive common strip and uniform coefficient bounds; no new pointwise assertion is assumed from their Hilbert estimates. For the scalar notation below, with $p_r = z + 2\pi r$, set

$$a_r = \prod_\nu M^{-1} \sum_{s \in I_M} e^{ip_{r,\nu} s/M}, \qquad \omega_{r,\mu} = a_r M^{-1} \sum_{t=0}^{M-1} e^{ip_{r,\mu} t/M},$$

$$R_{s,2} = \sum_{r \neq 0} \frac{a_r a_r^-}{P_r^2}, \qquad D_{s,2} = a_0 a_0^- + P_0^2 R_{s,2}.$$

The symbols P_r and the negative-argument convention are recalled in the next section. These are scalar denominators; the subscript $s,2$ is a name and does not designate a free matrix index. The homogeneous source value of the one-form penalty is $a = 1$; a fixed compact positive interval is allowed throughout.

2 Pointwise regularity for arbitrary fine forcing

The full flat inverse also obeys pointwise estimates. These estimates concern arbitrary bounded fine forcing, without assuming that it is a coarse lift. Retain fine spacing $\eta = M^{-1}$, odd M, unit coarse cells, the physical fine measure η^d, and

$$\mathcal{A}_a = d^* d + DRD^* + Q^* a Q, \qquad G_0 = \mathcal{A}_a^{-1}, \qquad \mathcal{L} = D^* D. \tag{1}$$

Here d is the fine curl, D the scalar gradient, $Q = Q_M^{\mathrm{st}}$, and R is the matched homogeneous flat scalar source projection. The penalty lies in a fixed interval $0 < a_- \leq a \leq a_+ < \infty$; the normalized source value $a = 1$ is included. The discussion applies to the homogeneous infinite lattice and compatible periodic regulators.

For a fixed coarse-cell patch U, define the discrete pointwise norm

$$\|u\|_{1,\alpha;U} = \|u\|_{\infty;U} + \sum_{\mu} \left\| D_\mu^+ u \right\|_{\infty;U} + \sum_{\mu} \sup_{\substack{x,y \in U \\ x \neq y}} \frac{|D_\mu^+ u(x) - D_\mu^+ u(y)|}{|x-y|_1^\alpha}. \tag{2}$$

The points are actual fine-grid points and a fixed collar supplies the differences. Enlarging the patch by finitely many neighboring coarse cells changes only constants. Backward differences obey the same bounds.

Theorem 2.1. *For each $0 < \alpha < 1$ there are $C_\alpha, \rho > 0$, independent of odd M and compatible periodic volume, such that the block from forcing in coarse cell m to a fixed collar of coarse cell n satisfies*

$$\|G_0(n,m)\|_{\ell^\infty \to C^{1,\alpha}} \leq C_\alpha e^{-\rho |n-m|_1}, \tag{3}$$

$$\|(\mathcal{L} G_0)(n,m)\|_{\ell^\infty \to \ell^\infty} \leq C e^{-\rho |n-m|_1}. \tag{4}$$

The first target norm is (2); $\mathcal{L}$ in the second estimate acts componentwise on the vector output. On tori use periodic coarse-cell distance. Weighted row and column block sums are uniformly bounded at every smaller positive exponent. Consequently bounded fine forcing produces uniformly bounded fields, first differences, locally α-Hölder first differences, and componentwise Laplacians.

The constants need not stay bounded as $\alpha \uparrow 1$. In fact a uniform Lipschitz estimate for the first differences, and a uniform bound for each individual second difference, are false in dimension $d \geq 2$; an example is given below. No source weak-norm identification is inferred merely from the theorem's terminology.

2.1 Reduction to a scalar high-mode inverse

Use the aliases $p_r = z + 2\pi r$, $r \in I_M^d$, with $I_M = \{-(M-1)/2, \ldots, (M-1)/2\}$, and write

$$u_{r,\mu}^{\pm} = M(e^{\pm i p_{r,\mu}/M} - 1), \qquad P_r = \sum_\mu u_{r,\mu}^+ u_{r,\mu}^-, \qquad \mathcal{F}_M(z)_{sr} = e^{i(z+2\pi r)\cdot s/M}. \tag{5}$$

The common holomorphic chart from the full inverse theorem contains the real fundamental cube in the interior of $|\operatorname{Re} z_\mu| < K < 2\pi$. For real z, $\mathcal{F}_M$ is unitary from alias counting ℓ^2 to physical-cell L^2. On the complex chart it and its inverse have uniformly bounded L^2 norms. In particular a bounded fine-cell input has alias coefficients bounded in ℓ^2, and individually, by $C\,\|f\|_\infty$.

Recall the canceled inverse $G_0 = F + \beta E_a^{-1}\beta^-$. Separate the scalar orientation-diagonal multiplier

$$T_r = P_r^{-1} \quad (r \neq 0), \qquad T_0 = 0. \tag{6}$$

The difference $F - T$ is a fixed sum of rank-one operators per vector orientation: its central block, two mixed blocks, and one high/high term. Its high vectors are $\omega^-_{r,\mu}/P_r$ and the analytic transpose partners. The remaining scalar coefficients are uniformly bounded and holomorphic. Thus the only full-rank pointwise problem is (6). Minus signs on multipliers throughout this argument denote evaluation at negative full momentum, with the analytic transpose; they do not conjugate a complex variable.

2.2 Weighted heat estimates with complex Bloch twist

Let $p_s(n)$ be the continuous-time nearest-neighbor heat kernel on $\mathbb{Z}$ with generator $f(n+1) + f(n-1) - 2f(n)$. Its Poisson-process representation gives

$$\sum_n e^{tn} p_s(n) = e^{2s(\cosh t - 1)}, \qquad \sum_n p_s(n) = 1. \tag{7}$$

It is symmetric and decreasing for $n \geq 0$. For completeness, set $q_n = p_s(n) - p_s(n+1)$ on $n \geq 0$. Then

$$q_0' = q_1 - 3q_0, \qquad q_n' = q_{n-1} + q_{n+1} - 2q_n \quad (n \geq 1),$$

with nonnegative initial data. This bounded cooperative system preserves nonnegativity, as follows by shifting its generator to a matrix with nonnegative entries and expanding the exponential. Also Fourier inversion gives

$$p_s(0) = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-4s \sin^2(t/2)}\, dt \leq C(1+s)^{-1/2}. \tag{8}$$

For $0 \leq r \leq r_0$, summation by parts therefore yields the exact weighted variation identity

$$\begin{aligned} \sum_n e^{r|n|} |p_s(n+1) - p_s(n)| &= (1 + e^r) \left[p_s(0) + (1 - e^{-r}) \sum_{n \geq 1} e^{rn} p_s(n) \right] \\ &\leq C \left[(1+s)^{-1/2} + r e^{2s(\cosh r - 1)} \right]. \end{aligned} \tag{9}$$

Take $s = M^2 t$ and $r = \delta/M$, where δ is a fixed small constant. Multiplying the difference by M gives

$$\sum_n e^{\delta|n|/M} |D_\eta p_{M^2 t}(n)| \leq C(t^{-1/2} + \delta) e^{C\delta^2 t}. \tag{10}$$

For $t < \eta^2$ the trivial M bound is smaller than $t^{-1/2}$. The weight is submultiplicative. Apply (10) twice to $p_s = p_{s/2} * p_{s/2}$ to obtain the second-difference bound. Tensor products then give, for any specified composition of k fine forward/backward differences in d dimensions,

$$\sum_{n \in \mathbb{Z}^d} e^{\delta|n|_1/M} |D_\eta^k p_{M^2 t}(n)| \leq C t^{-k/2}, \qquad 0 < t \leq 1, \quad k = 0, 1, 2. \tag{11}$$

The displayed p is a probability kernel. If it is instead written as a density against physical fine volume, the density factor M^d cancels the summation factor M^{-d}.

The reconstructed Bloch heat operator has the exact image-sum kernel

$$H_z(t;s,s') = \sum_{k\in\mathbb{Z}^d} e^{-iz\cdot k} p_{M^2t}(s-s'+kM). \tag{12}$$

This convention uses ordinary summation over the fine-cell input index s'. The formula follows by finite Fourier inversion and continues to complex z by the exponential moment bound. For s in a fixed output collar and s' in the input cell,

$$|e^{-iz\cdot k}| \le Ce^{\delta|s-s'+kM|_1/M}.$$

Thus (11), including the corresponding output differences of the image sum, proves

$$\left\|D_\eta^k H_z(t)\right\|_{\infty\to\infty} \le Ct^{-k/2}, \qquad 0<t\le 1, \quad k=0,1,2. \tag{13}$$

The difference between first-gradient rows at fine points x, y is bounded both by $Ct^{-1/2}$ and, by telescoping, by $C|x-y|_1 t^{-1}$. Interpolating these two elementary bounds gives

$$\|H_z(t)\|_{\infty\to C^{1,\alpha}} \le C_\alpha t^{-(1+\alpha)/2}, \qquad 0<t\le 1. \tag{14}$$

Subtracting the central heat mode changes this only by a bounded term, since P_0, u_0 and its plane-wave profile are uniformly bounded on the chart. For $t \ge 1$, the high-alias bound $\operatorname{Re} P_r \ge c|r|^2$ and the individual input-coefficient bound instead give

$$\begin{aligned}\left\|H_z(t) - e^{-tP_0}\Pi_0(z)\right\|_{\infty\to C^{1,\alpha}} \\ \le C\sum_{r\neq 0}(1+|r|)^{1+\alpha}e^{-ct|r|^2} \le C_\alpha e^{-ct/2}.\end{aligned} \tag{15}$$

The high-mode inverse is therefore the convergent operator integral

$$T(z) = \int_0^\infty [H_z(t) - e^{-tP_0}\Pi_0(z)]\,dt. \tag{16}$$

The small-time integral is finite because $\int_0^1 t^{-(1+\alpha)/2}\,dt = 2/(1-\alpha)$. This proves a uniform holomorphic $\ell^\infty \to C^{1,\alpha}$ bound for T for every $\alpha < 1$.

2.3 Pointwise bounds for the rank corrections

The centered scalar and straight-vector averaging multipliers satisfy

$$|a_r^\pm| \le Cw(r), \qquad |\omega_{r,\mu}^\pm| \le \frac{Cw(r)}{1+|r_\mu|}, \qquad w(r) = \prod_\nu (1+|r_\nu|)^{-1}. \tag{17}$$

Indeed, for a nonzero component, the numerator of $\sin((z_\nu+2\pi r_\nu)/2)/[M\sin((z_\nu+2\pi r_\nu)/(2M))]$ is uniformly bounded and its denominator has modulus at least $c|r_\nu|$ on the enlarged chart. The zero component is bounded directly. The vector multiplier has one extra longitudinal factor. This includes aliases at the edge of the fine zone.

On a dyadic maximum-norm shell,

$$\sum_{2^j\le|r|_\infty<2^{j+1}} w(r) \le C_d(j+1)^{d-1}. \tag{18}$$

Choose a coordinate at the maximal scale. Its reciprocal weight sums to $O(1)$ over that shell, and each other coordinate contributes $O(j+1)$. Summing over the possible maximal coordinates proves (18). Consequently

$$\sum_{r\neq 0}(1+|r|)^{1+\alpha}\left|\frac{\omega^-_{r,\mu}}{P_r}\right| \le C\sum_{r\neq 0} w(r)(1+|r|)^{\alpha-1} \le C_\alpha. \tag{19}$$

Absolute Fourier summation now bounds the reconstructed high rank vectors in $C^{1,\alpha}$. Their central partners have the same bound. The Woodbury vector has

$$|\beta_{r,\mu}| \le C\left(\frac{|a^-_r|}{(1+|r|)^3}+\frac{|\omega^-_{r,\mu}|}{(1+|r|)^2}\right), \qquad r\neq 0, \tag{20}$$

so its output profile satisfies the same weighted sum. Its canceled central component is bounded directly.

The input rows of every rank term have uniformly bounded alias ℓ^2 norm. By the physical-cell Fourier normalization,

$$|\text{input row}(f)| \le C\,\|f\|_{L^2(\text{physical cell})} \le C\,\|f\|_\infty .$$

The cell has physical volume one; there is no factor $M^{d/2}$. Combining these row estimates with (19)–(20) controls all rank corrections. The full reconstructed G_0 therefore has the required uniform holomorphic pointwise bound on the common chart.

2.4 The componentwise Laplacian

For the scalar term, $\mathcal{L}T = I - \Pi_0$ is bounded on ℓ^∞. Applying $\mathcal{L}$ to a high rank vector $\omega^-_{r,\mu}/P_r$ gives $\omega^-_{r,\mu}$. Its full reconstruction is a column of the physical adjoint average, minus its bounded central mode. Explicitly, for $x = n + s/M$, $M = 2h+1$,

$$(Q^*B)_\mu(x) = b_s B_\mu(n) + (1-b_s)B_\mu(n-e_\mu), \qquad b_s = \frac{s_\mu + h + 1}{M}. \tag{21}$$

These weights are nonnegative and sum to one. The Bloch profile is $b_s + (1-b_s)e^{-iz_\mu}$, uniformly bounded on the complex chart and any fixed collar. An absolute alias sum of ω is unnecessary and would introduce a spurious logarithmic loss.

For the Woodbury vector, its high-alias formula gives

$$(\mathcal{L}\beta)_{r,\mu} = \frac{P_0 u^+_{r,\mu} a^-_r}{P_r D_{s,2}} - \omega^-_{r,\mu} k_\mu d^+_\mu . \tag{22}$$

The first term is $O(w(r)/(1+|r|))$, absolutely summable by (18). The second is a bounded coefficient times the profile (21), with its central subtraction. The canceled central Laplacian component is bounded. The same input row estimates as before now prove a uniform holomorphic $\ell^\infty \to \ell^\infty$ bound for $\mathcal{L}G_0$. This argument supplies neither Hölder continuity of its Laplacian nor bounds on all separate second differences. In particular the Q^* profile can have transverse block jumps.

2.5 Gluing, exponential decay, and bounded forcing

The whole reconstructed inverse and its fine derivatives are periodic in coarse momentum. The full real symbol is invariant under the alias relabeling induced by $z \mapsto z + 2\pi n$; the within-cell integer phase cancels wrapping of the alias index. The enlarged holomorphic charts agree on their real overlaps

and hence glue. Individual separated central/high terms need not be periodic; they were used only to bound the whole symbol on each chart.

Output collars consist of finitely many neighboring-cell copies, with bounded holomorphic factors $e^{iz\cdot n}$. Thus the common strip bounds hold as operator bounds into the discrete norm (2). For each fixed M these are finite-dimensional Banach spaces, so entrywise holomorphy and the proved uniform norms give Banach-valued contour shifting. This proves (3)–(4) without a growing-dimension factor.

In infinite volume the resulting absolutely summable blocks define the inverse on bounded forcing and agree with the Hilbert inverse on finitely supported fine forcing. The inverse identity extends through absolute convergence of the exponentially summable operator products at each fixed M. The source projection is not local, but $R = \mathcal{L}Z\mathcal{L}$ has exponentially summable cell L^2 blocks. Finite-cell norm equivalence makes their pointwise blocks summable for each fixed M. That last identity argument need not be uniform in M: the pointwise inverse bound was proved uniformly by the heat and rank estimates above.

Finite compatible tori sample the same symbol, giving exact periodizations and the same bounds in periodic distance, including short periods and multiplicities. Summing the cell kernels bounds the field and gradient globally and the gradient Hölder seminorm on every fixed collar. Nearby points lie in one enlarged collar; farther points use twice the gradient supremum. This also gives the global discrete Hölder bound, with periodic physical distance on tori.

2.6 A sharp obstruction to the Lipschitz endpoint

For $d \geq 2$, take the compatible torus consisting of one coarse cell. At its sole coarse momentum, zero, the full inverse is a^{-1} on the central constant mode and P_r^{-1} on every high alias. Thus on fine mean-zero forcing it is exactly the scalar periodic inverse Laplacian, independently in each vector component.

Choose a smooth cutoff χ supported in a small ball inside the unit continuum torus and equal to one near its center. Set

$$u_\varepsilon(x) = \chi(x) x_1 x_2 \, \frac{1}{2} \log(|x|^2 + \varepsilon^2). \tag{23}$$

For $P = x_1 x_2$ and $f(r) = \frac{1}{2}\log(r^2 + \varepsilon^2)$, harmonicity and homogeneity of P give

$$\Delta(Pf) = P\left(f'' + \frac{d+3}{r} f'\right) = P \frac{(d+2)r^2 + (d+4)\varepsilon^2}{(r^2+\varepsilon^2)^2}. \tag{24}$$

Since $|P| \leq r^2/2$, this is bounded independently of ε. The cutoff-region derivatives are also uniformly bounded because that region stays away from zero. However

$$\partial_1 \partial_2 u_\varepsilon(0) = \log \varepsilon. \tag{25}$$

For fixed $\varepsilon > 0$, sample the smooth periodic function on increasingly fine odd grids and subtract its exact discrete mean. Define $f_{M,\varepsilon} = \mathcal{L}_\eta u_{M,\varepsilon}$. Its mean is zero, and Taylor convergence gives $\|f_{M,\varepsilon}\|_\infty \leq C + 1$ for sufficiently large M, depending on ε. The full inverse returns exactly $u_{M,\varepsilon}$. Its mixed second fine difference at the origin converges to (25). Choosing small ε and then large M proves that bounded forcing can produce arbitrarily large mixed second differences. A uniform Lipschitz bound on first differences would bound these differences and is therefore impossible. This does not contradict (4), which controls their Laplacian sum rather than each second difference separately.

2.7 Source scope

The theorem supplies arbitrary-forcing pointwise estimates for the matched homogeneous flat inverse, including its normalized $a = 1$ case. It does not by itself identify every numerical factor or term in the

source's weak norms, establish nonflat estimates, handle general retained-history domains, or control the complete interacting current carrier. Those applications require exact norm and coordinate checks. The false endpoint above must not enter such a match through an implicit assertion that $C^{1,\alpha}$ controls all second derivatives.

3 The full flat current in the pointwise weak norms

We now transfer the pointwise estimates to arbitrary bounded complex fine currents. The background remains exactly flat. For the corrected nonlinear constraint term, refinement is $M = L^j$ with fixed odd L, as in the independently audited fixed-step quadratic-composition theorem. The result does not use a pointwise bound for the projection Π on arbitrary bounded fields and does not differentiate the current.

3.1 The source norms and spatial weights

At homogeneous $\ell = 1$ and background $U = 1$, deposited Raw I, lines 12740–12752, defines the equivalent norms

$$N(A) = \max\{\|A\|_\infty, \|\nabla A\|_\infty\}, \qquad N_\beta(A) = \max\{N(A), [\nabla A]_\beta\}, \quad 0 \le \beta \le \beta_0 < 1. \tag{26}$$

Here ∇ collects the first fine differences of a vector field; the Hölder member can be taken on fixed coarse-cell collars. The pointwise estimates of the preceding section also imply its global discrete version. There is no Laplacian in N_β. The source explicitly excludes the Laplacian member for the current inverse at Raw I, lines 13759–13765.

Let $\varpi(n) > 0$ be a cell weight satisfying

$$\frac{\varpi(n)}{\varpi(m)} \le e^{\kappa \operatorname{dist}(n,m)}. \tag{27}$$

The distance is ordinary or periodic cell distance. Define $\|f\|_{\infty,\varpi}$, N_ϖ, and $N_{\beta,\varpi}$ by the corresponding weighted supremum over cells and fixed collars. Neighboring weights have uniformly bounded ratios. In particular $\varpi(n) = e^{\kappa \operatorname{dist}(n,m_0)}$ is allowed. These norms are applied to complete fields. We never sharply cut a differentiable field into cells before applying D^*, which would introduce artificial fine-scale boundary derivatives.

Retain

$$\mathcal{L} = D^*D, \qquad Z = G'RG', \qquad \Lambda = Z\mathcal{L}D^* = G'RD^*, \qquad \Pi = I - D\Lambda. \tag{28}$$

Stars denote physical adjoints on the real slice and their fixed analytic bilinear continuations for complex parameters. The penalized baseline is $G_0 = (d^*d + DRD^* + Q^*aQ)^{-1}$, with matched unit-scale value $a = 1$.

3.2 The scalar smoothing needed for the current

For a common small $\kappa > 0$, the scalar inverse supplies constants independent of M and compatible volume such that

$$\|Z\mathcal{L}f\|_{\infty,\varpi} \le c_0 \|f\|_{\infty,\varpi}, \qquad \|DZ\mathcal{L}f\|_{\infty,\varpi} \le c_1 \|f\|_{\infty,\varpi}, \tag{29}$$

$$N_{\beta,\varpi}(DZf) \le z_1 \|f\|_{\infty,\varpi}, \qquad N_{\beta,\varpi}(DZD^*f) \le z_2 \|f\|_{\infty,\varpi}. \tag{30}$$

Constants can be chosen at the fixed ceiling β_0, uniformly for $0 \le \beta \le \beta_0$.

Here is the pointwise proof of these additional scalar facts. The canceled alias matrix of Z is the high diagonal P_r^{-2} plus a fixed number of rank terms whose high vectors are a_r^-/P_r^2 and their transpose partners. Its central entries have no massless pole. The high diagonal has the heat representation

$$\int_0^\infty t\,[H_z(t) - e^{-tP_0}\Pi_0(z)]\,dt. \tag{31}$$

The weighted heat derivative bounds extend to every fixed integer order k: split the semigroup into k equal-time factors, put one fine difference on each, and use weighted L^1 convolution. Thus the small-time bound is $C_k t^{-k/2}$. An additional β-Hölder output difference costs $t^{-\beta/2}$ by the same interpolation used for the baseline inverse. Input derivatives are derivatives of the kernel, or discrete summation by parts, and do not require derivatives of the forcing. The relevant integral is

$$\int_0^1 t^{1-(m+\beta)/2}\,dt < \infty \quad \text{if } m+\beta < 4.$$

For $Z\mathcal{L}, DZ\mathcal{L}$ the derivative orders are two and three. For the gradient Hölder members of DZ, DZD^* they are two and three, respectively, before the extra β. Large times use the high-alias gap.

For the rank terms, an input row with the required derivatives has uniform alias ℓ^2 norm. Output vectors obey $|a_r/P_r^2| \le Cw(r)/(1+|r|)^4$, where $w(r) = \prod_\nu (1+|r_\nu|)^{-1}$ and its dyadic shell sum is $O((j+1)^{d-1})$. Every output derivative/Hölder order strictly below four is therefore absolutely summable. Physical-cell L^2 input rows act on bounded forcing without a dimension factor. The same periodic holomorphic gluing as for G_0 gives exponential cell kernels and hence (29)–(30). Finite periodic sampling preserves the bounds.

It follows, using the actual derivative of the complete field, that

$$\|\Lambda A\|_{\infty,\varpi} \le l_0 N_\varpi(A), \qquad \|D\Lambda A\|_{\infty,\varpi} \le l_1 N_\varpi(A), \qquad \|\Pi A\|_{\infty,\varpi} \le (1+l_1) N_\varpi(A). \tag{32}$$

No assertion that $D\Lambda$ or Π is bounded on unrestricted ℓ^∞ input is used.

3.3 Both current terms map N to N_β

The matched local auxiliary factorization is

$$\Theta'_J = \mathrm{j}_J^*\Lambda + \Lambda^*\mathrm{j}_J + \Lambda^* V_J D\Lambda + (D\Lambda)^* V_J^*\Lambda. \tag{33}$$

The finite-incidence coefficient operators j_J, V_J and their analytic transposes have weighted pointwise norms at most $c_j\|J\|_\infty, c_v\|J\|_\infty$. They contain no derivative of J. The exact scalar identity $G_0\Lambda^* = DZ$ gives

$$G_0\Theta'_J = G_0\mathrm{j}_J^*\Lambda + DZ\mathrm{j}_J + DZV_J D\Lambda + DZD^*V_J^*\Lambda. \tag{34}$$

Let g_β bound G_0 from weighted bounded forcing to $N_{\beta,\varpi}$. Then

$$N_{\beta,\varpi}(G_0\Theta'_J A) \le C_{\mathrm{aux}}\|J\|_\infty N_\varpi(A), \tag{35}$$

where one possible constant is

$$C_{\mathrm{aux}} = g_\beta c_j l_0 + z_1 c_j + z_1 c_v l_1 + z_2 c_v l_0.$$

This is the homogeneous normalized source (J1), including weights. Its input requires N only, not Hölder regularity of A or J.

For the full constraint term, write

$$\Theta_J = T_{v_J}, \qquad v_J = E_J^T J, \qquad \langle A, T_v B\rangle_\eta = 2\langle C_j^{(2)}(A,B), v\rangle_1, \tag{36}$$

where E_J is the auxiliary normalized extension and $C_j^{(2)}$ is the quadratic jet of the actual corrected fixed-L composition. The previously proved auxiliary cell-Schur estimate gives, analytically, $\|v_J\|_\infty \le C_v \|J\|_\infty$ throughout its current ball.

The direct pointwise estimate for T_v uses the same physical factor as its Hilbert proof. At a step of fine spacing h, the quadratic coefficient is $(h/L)C_L$. Its finite coefficient tensor, paired with coarse measure and converted to the fine physical adjoint, has an absolute row sum at most $c_L h \|v\|_\infty$. The factor L^d in this conversion is fixed. The quadratic chain rule expresses the full operator as

$$T_v = \sum_{t=0}^{j-1} Q_{0:t}^* T_{h_t, v_t} Q_{0:t}, \qquad h_t = L^{t-j}, \qquad v_t = Q_{t+1:j}^* v. \tag{37}$$

Both flat Q and its physical adjoint Q^* have nonnegative row coefficients summing to one. They are ℓ^∞ contractions, including the covering-space multiplicities on short tori. Thus

$$\|T_v\|_{\infty\to\infty} \le \frac{c_L}{L-1} \|v\|_\infty .$$

The composition has fixed unit-cell range R_*, since its reading radii sum over the same geometric series. Consequently

$$\|T_v f\|_{\infty,\varpi} \le \tau_\infty \|v\|_\infty \|f\|_{\infty,\varpi} . \tag{38}$$

The constant includes $e^{\kappa R_*}$. This is a coefficient-row argument, not a conversion of a final-cell Hilbert norm into a pointwise norm.

The identity

$$G_0 \Pi^* = G_0 - DZD^* \tag{39}$$

now places every smoothing operator outside the bounded current coefficients. Combining (32), (38), and (39) gives

$$N_{\beta,\varpi}(G_0 \Pi^* \Theta_J \Pi A) \le C_{\mathrm{con}} \|J\|_\infty N_\varpi(A), \qquad C_{\mathrm{con}} = (g_\beta + z_2)\tau_\infty C_v (1 + l_1). \tag{40}$$

In particular no derivative of J has appeared.

There is also a constrained baseline estimate. Put

$$\Gamma = G_0 - H_{\mathrm{src}} q_{\mathrm{aux}}^{-1} H_{\mathrm{src}}^*, \qquad q_{\mathrm{aux}} = q_M + aI. \tag{41}$$

The extension H_{src} is pointwise local into N_β; its transpose maps bounded fine fields to bounded coarse fields by the physical-cell Schur bound; and $q_{\mathrm{aux}}^{-1} = QG_0Q^*$ is a local bounded coarse operator. Hence Γ maps weighted bounded forcing to N_β and

$$N_{\beta,\varpi}(\Gamma \Theta_J A) \le C \|J\|_\infty \|A\|_{\infty,\varpi} . \tag{42}$$

3.4 A common pointwise current neighborhood

Theorem 3.1. *Let ε_H be the established full-current Hilbert/Schur radius, which is contained in the auxiliary radius. Define*

$$C_* = C_{\mathrm{aux}} + C_{\mathrm{con}}, \qquad \varepsilon_{\mathrm{pt}} = \min\{\varepsilon_H, (2C_*)^{-1}\}, \tag{43}$$

increasing C_* by fixed norm-convention factors if necessary. For $\|J\|_\infty < \varepsilon_{\mathrm{pt}}$, put

$$\Psi_J = \Theta_J' + \Pi^* \Theta_J \Pi, \qquad K_J = G_0 \Psi_J. \tag{44}$$

The actual full homogeneous flat inverse is the convergent series

$$G_1^J f = (I - K_J)^{-1} G_0 f = \sum_{n \geq 0} K_J^n G_0 f. \tag{45}$$

It is holomorphic from the complex bounded-current ball to operators from bounded fine forcing into N_β, uniformly for $0 \leq \beta \leq \beta_0$. With g_N the baseline bounded-forcing-to-N constant, it obeys

$$N_\varpi(G_1^J f) \leq 2 g_N \|f\|_{\infty,\varpi}, \tag{46}$$

$$N_{\beta,\varpi}(G_1^J f) \leq (g_\beta + 2 C_* \varepsilon_{\mathrm{pt}} g_N) \|f\|_{\infty,\varpi}. \tag{47}$$

All constants are independent of $M = L^j$ and compatible volume. For forcing supported in cell m, choosing $\varpi(n) = e^{\kappa \operatorname{dist}(n,m)}$ gives exponential coarse-cell locality in the pointwise target norms. The auxiliary inverse is the same result with only Θ'_J retained.

Proof. Equations (35) and (40) give

$$N_{\beta,\varpi}(K_J A) \leq C_* \|J\|_\infty N_\varpi(A).$$

The target dominates N_ϖ, so the ratio is below $1/2$ on (43). The series converges in weighted N. For every term after the zeroth, apply the stronger N-to-N_β bound to its outside K_J and the N contraction to the remaining factors. Summing gives (46)–(47). The coefficient $\Theta_J = T_{E_J^T J}$ is nonlinear in J but is analytic on the auxiliary ball with the proved linear norm bound. Uniform convergence therefore gives Banach-valued holomorphy; no linear-current differentiation formula is presumed for the full family.

For forcing in one cell, the exponentially weighted solution is in physical L^2 on the infinite lattice. It consequently equals the independently constructed Hilbert inverse on that input. Equality for bounded forcing follows by the absolutely summable pointwise cell kernels and the fixed-M summability of the original operator products. The finite-volume proof is identical with periodic distance. The minimum with ε_H ensures that both constructions use the same actual source inverse. □

The normalized extensions inherit the same bounds: $E_1^J = G_1^J Q^* q_1^J$ and $E_J = G_\pi^J Q^* q_J$. Here Q^* maps bounded coarse fields to bounded fine fields, and the established coarse Schur bounds make q_1^J, q_J bounded on weighted coarse ℓ^∞. Thus their output N_β norms are uniform on the common ball.

3.5 What has and has not matched the source

The input and output norms in (35) and (40) are the explicit homogeneous flat N, N_β norms, up to fixed equivalent conventions for localized Hölder seminorms. The current has only a supremum bound. These estimates and (47) supply the bounded-forcing members of the current program and their weighted versions.

No current Laplacian estimate is asserted. The source expressly excludes that member, and the Laplacian of DZD^* is an order-zero operator whose bounded-input pointwise norm has not been supplied. The following sections supply the one-form right-divergence members using complete weighted Hölder fields and smooth physical cutoffs, then treat the distinct scalar G' under its stated penalty hypothesis.

The fixed-step corrected-average source identification remains the one audited from the deposited transcriptions. These results are homogeneous and flat. They do not establish nonflat source stability, mixed retained-domain estimates or an interacting Yang–Mills ultraviolet limit. Companion Notebook 13 constructs a uniform analytic chart for the full fixed-reference nonlinear constraint; its moving-background identification is a further question.

4 Right-divergence data and the constrained current inverse

The remaining right-divergence estimates require cancellation on Hölder input. We isolate that cancellation in a translation-invariant massive lattice Green function. The argument applies to the full homogeneous flat G_0 and its auxiliary and full current perturbations. It does not identify the distinct scalar inverse G' with G_0.

Use the admissible spatial weights and weak norms of the preceding section. Let $\|\lambda\|_{\gamma,\varpi}$ denote the weighted supremum of the fine field and its γ-Hölder seminorm on fixed cell collars. Fields are kept complete; no Hölder datum is multiplied by a sharp cell indicator.

Theorem 4.1. *For $0 \le \beta < \gamma < 1$ and a sufficiently small fixed spatial exponent κ, the homogeneous flat inverses satisfy*

$$N_{\beta,\varpi}(GD^*\lambda) \le C_{\beta,\gamma}\,\|\lambda\|_{\gamma,\varpi}\,, \qquad G \in \{G_0, G_\pi^J, G_1^J\}. \tag{48}$$

The current lies in the previously established common pointwise and Hilbert ball. Constants are independent of refinement and compatible periodic volume. Here D^* can be any specified physical fine divergence, with its fixed orientation contractions. Taking $\gamma = \beta + \varepsilon$, with fixed $0 < \varepsilon < 1 - \beta_0$, makes the constants uniform for $0 \le \beta \le \beta_0$. The source choice $\varepsilon = (1-\beta_0)/2$ is included.

This is a bound on the right-differentiated operator, not an estimate obtained by bounding $\|D^*\lambda\|_\infty$. The constant can grow as $\gamma \downarrow \beta$; the endpoint is not claimed. The relevant source members are described at deposited Raw I, lines 13601–13615 and 13759–13765.

4.1 A remainder that accepts arbitrary bounded data

Introduce the comparison operator

$$G_m = (\mathcal{L}+1)^{-1} I, \qquad R_m = G_0 - G_m. \tag{49}$$

It is an auxiliary decomposition, not a change of source precision. Shrink the common complex chart so that $P_0 + 1$ is uniformly nonzero; on the real chart it is at least one. The high denominators already have this property. The high scalar diagonal of R_m is

$$\frac{1}{P_r} - \frac{1}{P_r+1} = \int_0^\infty (1-e^{-t})e^{-tP_r}\,dt, \qquad r \ne 0. \tag{50}$$

The other terms are the finite-rank corrections of G_0 and the bounded central subtraction $-(P_0+1)^{-1}\Pi_0$.

With one input derivative, one output derivative, and an output β-Hölder increment, the small-time bound from the heat estimates is

$$(1-e^{-t})t^{-(2+\beta)/2} \le t^{-\beta/2}, \qquad 0 < t \le 1.$$

It is integrable. Lower output members are easier, and large times use the high-alias gap. In each rank term, the input derivative multiplies its row by one fine multiplier. The high row ω_s/P_s becomes $O(|\omega_s|/(1+|s|))$; the Woodbury row is bounded by $C(|a_s|/(1+|s|)^2 + |\omega_s|/(1+|s|))$. Their alias ℓ^2 norms are uniform. The output $C^{1,\beta}$ vectors retain the proved absolute alias sums for $\beta < 1$.

The full reconstructed remainder is periodic and holomorphic in the resulting pointwise operator norms. Chart gluing and contour shifting therefore give exponential coarse-cell kernels and

$$N_{\beta,\varpi}(R_m D^* f) \le C_\beta\,\|f\|_{\infty,\varpi}\,, \qquad 0 \le \beta < 1. \tag{51}$$

This remainder estimate genuinely accepts arbitrary bounded fine data. It requires no Hölder extension across a cell boundary: the right derivative has already been placed on its kernel. Compatible periodic sampling gives the same bound on tori.

4.2 Weighted heat moments down to the mesh scale

Let $p_t^\eta(v)$ be the full-lattice probability heat kernel of $\mathcal{L}$, with fine displacement $v \in \eta\mathbb{Z}^d$. For $0 < \gamma < 1$ and $k = 2, 3$, we have

$$\sum_v e^{\kappa|v|_1} |v|_1^\gamma |D_\eta^k p_t^\eta(v)| \le C t^{(\gamma-k)/2}, \qquad 0 < t \le 1. \tag{52}$$

The probability-kernel convention includes the physical normalization: if a kernel density is used instead, its η^{-d} factor is canceled by physical summation.

For $t \ge \eta^2$, apply the weighted heat variation estimate with physical exponential scale $\delta = \kappa + t^{-1/2}$. Its lattice parameter $\eta\delta$ lies in a fixed bounded interval. Semigroup splitting into k factors gives

$$\sum_v e^{\delta|v|_1} |D_\eta^k p_t^\eta(v)| \le C_k t^{-k/2}, \qquad \eta^2 \le t \le 1,$$

because $\delta^2 t$ is bounded. The inequality $|v|^\gamma \le C_\gamma t^{\gamma/2} e^{|v|/\sqrt{t}}$ proves (52) in this range.

For $0 < t < \eta^2$, expand the finite difference into its fixed list of translates:

$$D_\eta^k p_t^\eta(v) = \eta^{-k} \sum_j c_j p_t^\eta(v + \eta a_j), \qquad |a_j|_1 \le k, \quad \sum_j |c_j| \le C_k.$$

The dimensionless random walk has time $t/\eta^2 < 1$, so its exponential and fixed positive moments are uniformly bounded. The weighted moment above is at most $C\eta^{\gamma-k}$. Since $k > \gamma$ and $t < \eta^2$, $\eta^{\gamma-k} \le t^{(\gamma-k)/2}$, proving (52) also below the mesh time.

For $t \ge 1$, absorb $|v|^\gamma$ into a fixed additional exponential weight $e^{\delta_1|v|}$. The weighted heat estimates are bounded by $Ce^{c(\kappa+\delta_1)^2 t}$. Choose κ, δ_1 small enough that this is at most $Ce^{t/2}$. The massive factor e^{-t} leaves an integrable tail. The required zeroth and first heat derivatives obey the analogous estimates without the polynomial moment.

4.3 Cancellation on complete fields

Every nonzero difference of the translation-invariant heat kernel has exact zero row sum:

$$\sum_v D_\eta^k p_t^\eta(v) = 0 \qquad (k \ge 1). \tag{53}$$

The local weighted Hölder norm implies

$$\varpi(\mathrm{cell}\, x)|\lambda(y) - \lambda(x)| \le C \|\lambda\|_{\gamma,\varpi} e^{\kappa|x-y|_1} |x-y|_1^\gamma. \tag{54}$$

At bounded separation use a common fixed collar and neighboring weight ratios. At larger separation use the two supremum bounds; the polynomial factor is then bounded below and the cell-weight ratio is controlled by physical distance up to a fixed offset. No derivative of the weight is taken.

Subtract $\lambda(x)$ using (53). Equations (52)–(54) give

$$\left\| D_\eta^2 e^{-t\mathcal{L}} \lambda \right\|_{\infty,\varpi} \le C t^{-1+\gamma/2} \|\lambda\|_{\gamma,\varpi}, \tag{55}$$

$$\left\| D_\eta^3 e^{-t\mathcal{L}} \lambda \right\|_{\infty,\varpi} \le C t^{-3/2+\gamma/2} \|\lambda\|_{\gamma,\varpi}, \qquad 0 < t \le 1. \tag{56}$$

The difference between two second-derivative output values is bounded by twice (55), and by their physical separation times (56). Interpolating yields

$$[D_\eta^2 e^{-t\mathcal{L}} \lambda]_{\beta,\varpi} \le C t^{-1+(\gamma-\beta)/2} \|\lambda\|_{\gamma,\varpi}. \tag{57}$$

Its integral is finite, with

$$\int_0^1 t^{-1+(\gamma-\beta)/2}\,dt = \frac{2}{\gamma-\beta}. \tag{58}$$

For $\beta = 0$ the supremum estimate suffices. The field member of $G_m D^*\lambda$ uses only the first heat derivative, whose $t^{-1/2}$ bound is integrable on bounded input. The gradient and its Hölder member use (55)–(58). The large-time massive tail is integrable as above. Therefore

$$N_{\beta,\varpi}(G_m D^*\lambda) \le C_{\beta,\gamma}\,\|\lambda\|_{\gamma,\varpi}\,. \tag{59}$$

All singular-integral cancellation has occurred in this global translation-invariant calculation. On tori use the periodized heat kernel, or periodically extend the fields and weights to the cover. Periodic distance is no larger than cover distance, so the same moment bounds and multiplicities give identical constants.

4.4 Transfer to the actual current inverses

Adding (51) and (59) proves (48) for G_0. For either current family, the proved weak-space perturbation obeys

$$N_{\beta,\varpi}(K_J A) \le C_*\,\|J\|_\infty\, N_\varpi(A), \qquad C_*\,\|J\|_\infty < \tfrac{1}{2}.$$

Apply the same convergent series to the baseline datum:

$$G^J D^*\lambda = (I - K_J)^{-1} G_0 D^*\lambda. \tag{60}$$

The baseline N bound requires positive input Hölder regularity; the baseline N_β bound uses $\gamma > \beta$. Every later term is controlled by the established N-to-N_β perturbation bound. This proves the remaining assertions of (48). The larger Hölder-data constant does not shrink the current radius: it multiplies the initial datum, not the contraction. Holomorphy follows from the same Neumann series.

Localization is legitimate with a smooth physical cutoff ϕ of fixed support collar and bounded first derivative. The product difference identity gives $\|\phi\lambda\|_{\gamma,\varpi} \le C_\phi\,\|\lambda\|_{\gamma,\varpi;\,extcollar}$ without a power of η^{-1}. A source-centered exponential weight then gives decay away from the support. A sharp cell indicator would generally create $\eta^{-\gamma}$ Hölder jumps and is not permitted by this argument. This matches the use of localized cutoffs in the stated source member.

4.5 The constrained-current member

The source separately defines, at Raw I, line 13735,

$$\widetilde{G}^J = (I - \Gamma\Theta_J)^{-1}\Gamma, \qquad \Gamma = G_0 - H_{\mathrm{src}} q_{\mathrm{aux}}^{-1} H_{\mathrm{src}}^*. \tag{61}$$

The preceding section proved $Q\Gamma = 0$ and, for a uniform constant B_Γ, $N_{\beta,\varpi}(\Gamma f) \le B_\Gamma\,\|f\|_{\infty,\varpi}$. It also proved $\|\Theta_J f\|_{\infty,\varpi} \le \tau_\infty C_v\,\|J\|_\infty\,\|f\|_{\infty,\varpi}$. Decrease the common current radius to

$$\varepsilon_{\mathrm{all}} = \min\left\{\varepsilon_{\mathrm{pt}}, \frac{1}{2B_\Gamma\tau_\infty C_v}\right\}, \tag{62}$$

omitting the second restriction if its coefficient is zero. The series in (61) then converges in weighted pointwise norms and gives the source (J6) estimate

$$N_{\beta,\varpi}(\widetilde{G}^J f) \le 2B_\Gamma\,\|f\|_{\infty,\varpi}\,, \qquad Q\widetilde{G}^J = 0. \tag{63}$$

For the bound, estimate the outside Γ in N_β and sum the remaining supremum-norm geometric series. For the exact constraint, every term begins on the left with Γ, so applying Q gives zero term by term; the limit preserves the identity. The map is analytic, and source-centered weights give exponential coarse-cell locality. This statement keeps the source's constrained inverse distinct from G_π^J and G_1^J.

4.6 Remaining distinctions

The right-divergence theorem covers G_0, G_π^J, G_1^J for the stated Hölder input and also supplies the separate constrained current member (63). The deposited source also names a right-divergence estimate for the distinct scalar inverse G'. That member is not inferred by replacing G' with G_0. The following section gives its own scalar Woodbury adaptation for a penalty in a fixed positive interval; verifying any separately specified source penalty remains an application condition.

No current Laplacian estimate is asserted. The positive-loss argument above will next be strengthened at each fixed equal exponent in $(0, 1)$. The results are homogeneous and flat, with the corrected fixed-step constraint term already matched to the deposited transcription. They do not prove nonflat source stability, mixed retained-domain bounds, strict background locality, or an interacting Yang–Mills ultraviolet limit.

5 Equal-exponent Schauder bounds and the scalar inverse

The positive Hölder loss in the preceding right-divergence theorem is sufficient for the displayed source estimates. It is not necessary for a fixed exponent strictly between zero and one. The following refinement is useful for nonlinear elliptic equations, where an exponent loss at every iteration would not close a fixed point argument.

Theorem 5.1. *For every fixed $0 < \alpha < 1$, the homogeneous flat operators obey*

$$N_{\alpha,\varpi}(GD^*\lambda) \le C_\alpha \|\lambda\|_{\alpha,\varpi}, \qquad G \in \{G_0, G_\pi^J, G_1^J\}. \tag{64}$$

The spatial exponent is sufficiently small, and the current is in the common ball for the N-to-N_α estimates established above. Constants are uniform in refinement and compatible volume. There is no uniform assertion as α tends to zero or one.

Proof. Use $G_0 = G_m + R_m$ from (49). Equation (51) already controls $R_m D^*$ from bounded input into N_α for every $\alpha < 1$. Only the gradient Hölder member of $G_m D^*$ needs a new argument. The heat moment and exact row cancellation, with $\gamma = \alpha$, give

$$\left\|D^2 e^{-t\mathcal{L}}\lambda\right\|_{\infty,\varpi} \le Ct^{-1+\alpha/2}\|\lambda\|_{\alpha,\varpi}, \qquad \left\|D^3 e^{-t\mathcal{L}}\lambda\right\|_{\infty,\varpi} \le Ct^{-3/2+\alpha/2}\|\lambda\|_{\alpha,\varpi}$$

for $0 < t \le 1$, including times below the mesh scale. For two fine output sites in a fixed collar at distance $0 < r \le 1$, estimate their second-derivative difference by the first bound for $t < r^2$, and by r times the second for $t \ge r^2$. The finite path used for this difference stays in a fixed enlarged collar, so its weight comparison costs only a fixed constant. Consequently the difference after integrating the massive heat kernel is bounded, divided by the input norm, by

$$C\int_0^{r^2} t^{-1+\alpha/2}\,dt + Cr\int_{r^2}^1 t^{-3/2+\alpha/2}\,dt + Cr \le C\left(\frac{2}{\alpha} + \frac{2}{1-\alpha} + 1\right) r^\alpha. \tag{65}$$

The last term is the large-time massive tail, using its third derivative bound. Distances above one are controlled by the supremum bound. The latter uses $\int_0^1 t^{-1+\alpha/2}dt = 2/\alpha$ for the gradient and the integrable first heat derivative for the field. This proves the baseline assertion. It improves the preceding argument by splitting the time integral for each output separation before taking the Hölder supremum, rather than integrating a fixed Hölder operator norm down to zero time.

The current transfer is exactly (60): the baseline datum is in N_α, and the perturbation maps N to N_α with a strict contraction in N. Its radius does not depend on the initial datum constant. Periodic extension, complete weighted fields and smooth localization work as before. □

5.1 A separately normalized scalar statement

Let

$$G' = (\mathcal{L} + Q_0^* a' Q_0)^{-1}, \qquad 0 < a'_- \le a' \le a'_+ < \infty,$$

where the distinct scalar penalty ranges in a fixed compact positive interval. This hypothesis is explicit: no unprinted scale-dependent scalar penalty is assigned a value here. With the scalar aliases a_r, put

$$A = a_0 a_0^-, \quad R_{s,1} = \sum_{r \neq 0} \frac{a_r a_r^-}{P_r}, \quad \widehat{R}_{s,1} = a'^{-1} + R_{s,1}, \quad D_{s,1} = A + P_0 \widehat{R}_{s,1}. \tag{66}$$

The canceled scalar Woodbury formula is

$$\begin{aligned} G'_{00} &= \widehat{R}_{s,1}/D_{s,1}, \\ G'_{0s} &= -a_0^- a_s/(P_s D_{s,1}), & G'_{r0} &= -a_r^- a_0/(P_r D_{s,1}), \\ G'_{rs} &= \frac{\delta_{rs}}{P_r} - \frac{P_0 a_r^- a_s}{P_r P_s D_{s,1}}, & & r, s \neq 0. \end{aligned} \tag{67}$$

These formulas extend across $P_0 = 0$; no singular factor $1/P_0$ remains. They follow by the rank-one inverse identity away from $P_0 = 0$ and analytic continuation. On the real compact chart, $D_{s,1} \ge A \ge c > 0$. Alias Parseval bounds and the high-alias gap give uniform derivative bounds, so $D_{s,1}$ remains bounded away from zero on a common complex strip. The constants may depend on a'_-, a'_+.

Subtract the scalar massive inverse $G_m = (\mathcal{L} + 1)^{-1}$. The high diagonal has exactly the improved heat representation (50). All other terms in (67) have bounded central coefficients and high output vector a_r^-/P_r. This vector has uniform absolute alias summability after $1 + \alpha$ physical differences when $\alpha < 1$: the bound $|a_r| \le C \prod_\nu (1 + |r_\nu|)^{-1}$ gives a dyadic shell sum bounded by $C(1+n)^{d-1} 2^{-n(1-\alpha)}$. One right derivative leaves the input row bounded in alias ℓ^2, since $|u_r a_r / P_r| \le C|a_r|/(1 + |r|)$ for high aliases. The undifferentiated input rows have the same property. Uniform cell Fourier conversion, periodic gluing and contour shifting therefore give

$$N_{\alpha,\varpi}((G' - G_m) D^* f) \le C_\alpha \, \|f\|_{\infty,\varpi} \, . \tag{68}$$

The bounded central massive subtraction is included in this estimate. Only the complete reconstructed operator is periodized and shifted; the individual central pieces need not be periodic.

Applying the same massive cancellation proof establishes

$$N_{\alpha,\varpi}(G' D^* \lambda) \le C_\alpha \, \|\lambda\|_{\alpha,\varpi} \, , \qquad 0 < \alpha < 1. \tag{69}$$

Using instead the preceding positive-loss proof also gives $N_{\beta,\varpi}(G' D^* \lambda) \le C_{\beta,\gamma} \, \|\lambda\|_{\gamma,\varpi}$ for $0 \le \beta < \gamma < 1$. This proves the scalar analytic estimate for the stated penalty normalization. Application to a separately specified source penalty requires verifying its uniform positive interval. It neither replaces G' by the one-form G_0 nor identifies their gauge projections.

Equal Hölder exponents strictly between zero and one in (64) do not assert a Lipschitz-gradient endpoint or a second-difference bound on arbitrary bounded forcing. The earlier endpoint counterexample remains applicable to those other claims.

References

[1] Pinned Yang–Mills-LLM-experiment account, commit `e392ef45013a581664bab76b90867170b30fe10e`. The homogeneous weak norms are Raw I:12740–12770; the bounded current assumption is I:12951–12966; the distinct right-divergence members are I:13601–13615; the current assertions and exclusion of the current Laplacian member are I:13749–13804.

[2] Companion research proofs: Notebook 09, *The flat fine inverse and a current neighborhood*; Notebook 11, *The full flat constraint-current family*. Source operator matching is to the pinned deposited transcriptions. Original primary-paper arguments have not all been independently reverified.